

# Inference of Protein Secondary Structure with HMMs and BayesFlow

Simulation-Based Inference

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# Introduction & Modeling Idea

## The Problem

Predict per-position secondary structure from variable-length amino acid sequences.

## Two-State HMM

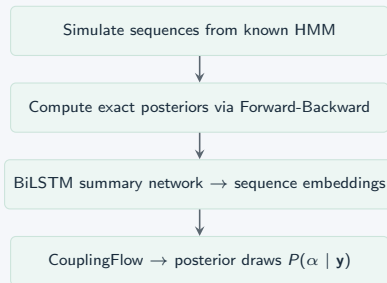
State 0 =  $\alpha$ -helix

State 1 = other ( $\beta$ -sheets & coils)

## Modeling Goal

Approximate  $P(S_t = \alpha \mid \mathbf{y})$  at every position  $t$  using **BayesFlow** neural posterior estimation.

## Overall Pipeline



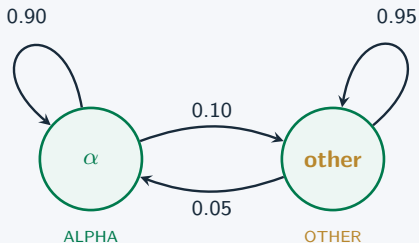
## Validation

Compare to experimental ground truth:

Human insulin **1A7F** (chain A: 21 aa, chain B: 29 aa).

# The 2-State Hidden Markov Model

## State Transitions



**Deterministic start:**  $P(S_1 = \text{other}) = 1.0$ ,  $P(S_1 = \alpha) = 0.0$

**Highly persistent:**  $\alpha$  stays  $\alpha$  (90%), other stays other (95%)

## Selected Emission Contrasts

| Amino acid   | $\alpha$ -helix | other | Role          |
|--------------|-----------------|-------|---------------|
| A (Alanine)  | 12%             | 6%    | Strong former |
| E (Glutamic) | 9%              | 5%    | Former        |
| P (Proline)  | 2%              | 6%    | Breaker       |
| G (Glycine)  | 4%              | 9%    | Breaker       |

Full  $20 \times 2$  table in written report.

## Prior Dynamics

**Stationary distribution:**

$$\pi(\alpha) = 0.05 / (0.05 + 0.10) = 1/3 \approx 33\%$$

$$\pi(\text{other}) \approx 67\%$$

Prior strongly favors **non-helical** states ( $\sim 67\%$ ).

# Data Generation & Analytical Targets

## Simulator

Sequential Markov sampling of hidden states.

Independent randomized emission draws.

Random sequence lengths: 10–60 aa.

**100% simulated** — no real proteins.

## Target Computation

Exact posteriors via Forward-Backward.

`hmmlearn.CategoricalHMM` — manually

configured, never `.fit()`.

## Verification

Targets verified against independent

pure-NumPy FB solver. Confirms:

$P(S_0 = \alpha) \equiv 0$  (position 0 = seq. position 1).

## On-demand Generation

`generate_dataset(n, ...)` produces  
any number of (sequence, posterior,  
states) tuples. Downstream code

chooses the split sizes.

## Variable-Length by Design

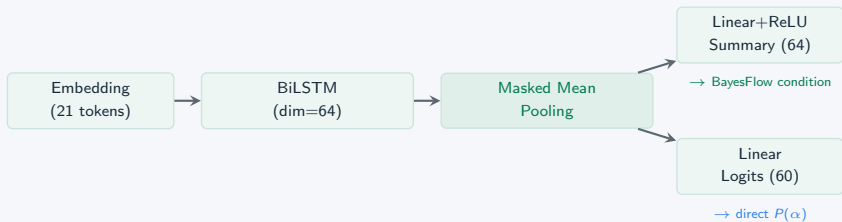
Sequences kept **ragged** (10–60 aa),  
each stored with its own length.  
The data layer does **not pad** — padding  
& masking are handled downstream by  
the summary network.

## Storage

`.npz` format: concatenated arrays +

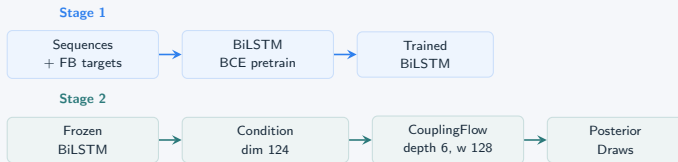
`lengths` vector. No object arrays.

# The Summary Network — BiLSTM



1. BiLSTM captures **sequential** dependencies — permutation-invariant nets fail here.
2. Padding token 20 & masked loss → handles variable-length sequences.
3. Two-stage: BiLSTM pretrained on BCE → frozen → CouplingFlow on top.

## Two-Stage Training & CouplingFlow



1. BiLSTM pretrained on masked BCE  $\rightarrow$  frozen  $\rightarrow$  condition (dim 124).
2. CouplingFlow: depth 6, hidden 128, affine, actnorm  $\rightarrow$  posterior draws.
3. **Model selection:** **not** flow val\_loss. Rank by masked MAE & correlation.

# Diagnostics: Adapted Simulation-Based Calibration

## Why Standard SBC Does Not Apply

**Standard SBC** (Talts et al., 2018):

$\theta \sim p(\theta)$  (draw parameter)

$x \sim p(x | \theta)$  (simulate data)

Rank  $\theta_{\text{true}}$  in posterior

→ Ranks must be **uniform**.

### The Problem

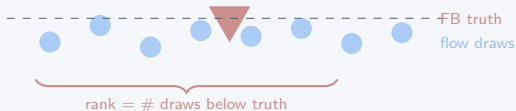
HMM parameters are **fixed**.

$P(\alpha | \mathbf{y})$  is a **deterministic** function of the sequence.

**No  $\theta$  to sample** from a prior.

→ Standard SBC is impossible.

## Our Adaptation: Position-wise SBC



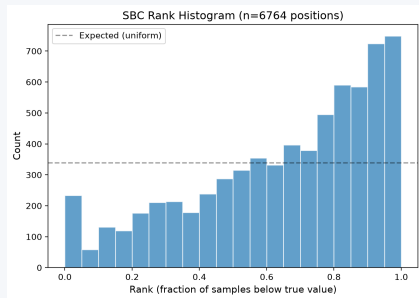
**Key idea:** Instead of ranking an unknown parameter, rank the **known** FB target within the flow's posterior draws at **each sequence position**.

**6,764 positions** tested → rank histogram, PIT ECDF, KS statistic.

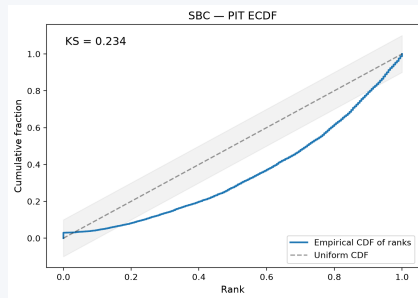
**Tests:** Does the flow's posterior **envelope** the true value uniformly?

≡ Is the posterior **calibrated per quantile**?

# Calibration Results — Adapted SBC Diagnostics



Rank histogram (left)



PIT ECDF with KS statistic (right)

6,764 positions

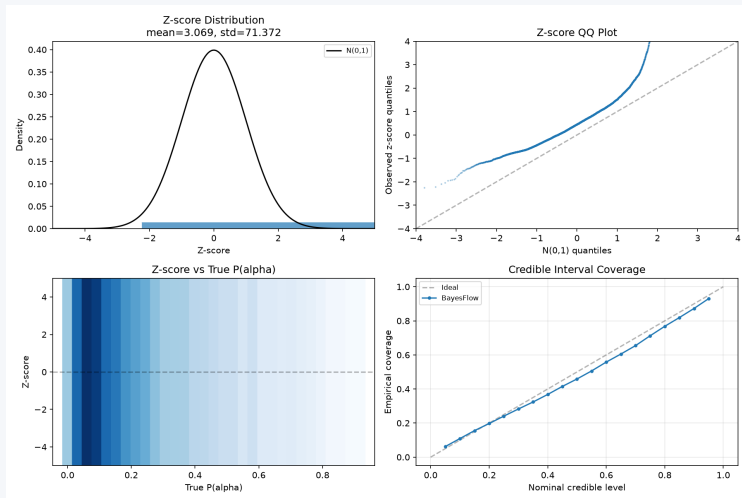
## Adapted SBC Results

- Mean Rank = **0.65** (Rightward skew)
- Only **27%** of ranks below 0.50
- KS Distance = **0.234** (Highly significant)

## The Spatial Cause

Position 0 (deterministic start) and padded boundaries are **exact zeros**. Normalizing flows are continuous — they struggle to represent **point masses**.

# Calibration Results — Z-Score & Coverage Diagnostics



## Key Metrics

Raw Z-std = **71.4**

(pos-0 degeneracy)

Trimmed (1.1%):

mean = **0.63**

std = **1.27**

Coverage:

**87% vs. 90%**

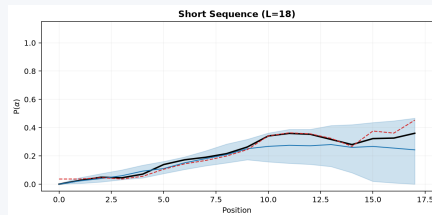
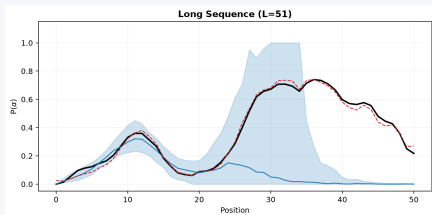
## Overconfidence

Biased mean (0.63)

+ std > 1.0 (1.27)

→ posteriors **too narrow** at boundaries.

# Synthetic Evaluation & Posterior Contraction



Black = True  $P(\alpha)$    Blue band = BayesFlow 90% CI   Red dashes = BiLSTM point

## Accuracy Metrics

| Model                     | MAE          | $r$          |
|---------------------------|--------------|--------------|
| Baseline (sequence-blind) | 0.162        | 0.200        |
| BayesFlow (median)        | 0.115        | 0.560        |
| <b>BiLSTM (point)</b>     | <b>0.060</b> | <b>0.988</b> |

6,764 positions. Both beat the naive baseline.

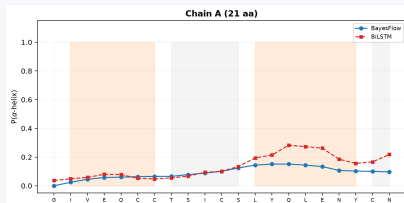
### Why BiLSTM Wins

$P(\alpha | \mathbf{y})$  is a **deterministic** function of the sequence. A direct network learns it with near-zero error. A flow models a residual density.

### Why We Still Use BayesFlow

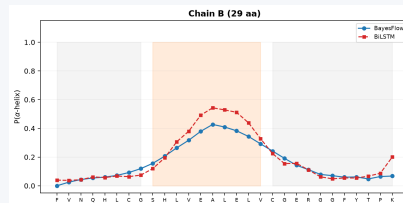
BiLSTM yields **no uncertainty**. BayesFlow provides full amortized posteriors  $\rightarrow$  credible intervals, SBC, z-scores, and coverage diagnostics.

# Real-World Validation — Human Insulin (1A7F)



Orange = Experimental Helix

Grey = Experimental Other



Blue = BayesFlow

Red = BiLSTM

## The Statistical Mechanism

### Chain B — Agreement ( $r=0.86$ )

34.5% true helix  $\approx \pi=33\%$  prior. Prior and evidence agree  $\rightarrow$  high accuracy.

### Chain A — Conflict ( $r=0.17$ )

61.9% true helix  $\gg 33\%$  prior. Severe prior-data conflict over only 21 residues.

*Both beat sequence-blind baseline (MAE=0.162,  $r=0.200$ ).*

## Key Proof — Model Limitation

### Exact FB also fails on Chain A

Running the **exact analytical Forward-Backward** solver on Chain A achieves only  $r \approx 0.40$ .

$\rightarrow$  The failure is a **model limitation** of the 2-state HMM, **not** a neural network issue.

No amount of training can exceed the HMM's ceiling.

# Limitations, Fixes & Take-Home Messages

## Limitation & Fix: The HMM

Expand to **3 states** ( $\alpha$ ,  $\beta$ , coil).

Treat transition probabilities as **inferable** parameters  $\theta$  under Beta priors.

→ BayesFlow would then recover a genuine posterior over  $\theta$ .

## Fix: The CouplingFlow

Pass targets through a **logit transformation**

before training → strictly bound all draws in  $(0, 1)$ .

## Take-Home Messages (TL;DR)

1. BiLSTM summary nets capture sequential dependencies where permutation-invariant nets fail.
2. BayesFlow successfully amortizes posterior estimation, enabling robust uncertainty quantification.
3. Real-world validation demonstrates the necessity of prior typicality checking — identify prior-data conflicts.

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Code:

[github.com/armin2080/](https://github.com/armin2080/)

[protein-secondary-structure-npe](#)